A Quantitative Discourse Analysis of Asian Workers in the US Historical Newspapers

Anonymous EACL submission

Abstract

This document is a supplement to the general instructions for *ACL authors. It contains instructions for using the LaTeX style file for EACL 2023. The document itself conforms to its own specifications, and is, therefore, an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers.

1 Introduction

Digitization of historical texts has opened up new opportunities for researchers to explore a large-scale corpus of historical texts with computational methods. Especially, the ready availability of datasets required for Natural Language Processing (NLP) research has welcomed researchers from other disciplines to NLP research (Park and Jeoung, 2022) and reduced the barriers to entry of NLP research. Taking this advantage, many researchers in diverse field, namely Sociology, History, English, and Information Science have applied NLP techniques to historical texts, such as books (Parulian et al., 2022), newspapers (Smith et al., 2013; Pedrazzini and McGillivray, 2022; Santin et al., 2016), and/or congressional records (Lin and Peng, 2022; Guldi, 2019) by the help of available archival metadata (Dobreski et al., 2019). Creating an interdisciplinary research space called *Digital Humanities*, there have been studies primarily focusing on the race problem in the United States. To introduce a few large-scale computational research, Soni et al. (2021) traced the semantic change of the word, for instance, when and which newspaper started to use the word with a new meaning and when and which newspapers adopted new semantic meaning of the word in abolitionist newspapers from the 19th century. Franzosi et al. (2012) analyzed racial violence, especially lynching performed by the White mob in the 19th century Georgia and presented quantitative narratives of the racial violence. These studies intersect the problem of historical racism and NLP research. However, despite the increasing attention toward the large scale text analysis on historical racism, there has been a gap of understanding the racism posed toward Asians in the United States. It is true that the major racial tension in the United States has been between White and Black. At the center of secession of the South and the creation of Confederacy, there is slavery problem. However, the racial tension between White and Asian has also been a part of the history of the United States. Throughout 19th and 20th century, Asian immigrant workers were derogatorily called *"coolies."* Britannica entry on *"coolie"* introduces the word as "pejorative European usage" to describe "an unskilled labourer or porter usually in or from the Far East hired for low or subsistence wages." [1] In seeking to fill the labor shortage in the United States due to the abolition, Chinese workers were recruited and the migration of Chinese workers arrived in the United States, beyond China-neighboring countries like India and Malaysia (Farley, 1968). Perceived to be a threat to White workers by replacing them with cheap labor (Rhoads, 2002), the tension culminated in prohibiting immigration of Chinese workers (Chinese Exclusion Act of 1882) (Lee, 2002). The problem of coolie exemplifies the extension of colonial and capitalist exploitation beyond Africa and sugarcoated the extended system as indentured migrant contract workers (Van Rossum, 2016). Therefore, studying coolie problem in the United States context supplements the historical study of racial tension mostly focused between White and Black and extends the racial conflict to include Asian population. To explore understudied topic of how Asian population was discriminated in the history of the United States, we present computational text analysis on how Asian workers are represented in the U.S. newspapers. We further

[1] https://www.britannica.com/money/topic/coolie-Asian-labourer

developed research questions as follows:

- RQ 1. How different are the semantic meaning of "coolie" in each State?
- RQ 2. What are the words over-represented in the newspapers between then-Confederate States and the rest?
- RQ 3. From where to where the story about coolie traveled?

To support open science and transparent data science, we publish the code used in this study at https://github.com/available-upon-accept.

2 Related Works

3 Methodology

3.1 Data collection

The data is collected from Chronicling America [2] API where digitized texts through optical character recognition (OCR) are accessible. When the word "coolie" was queried, API showed 124,511 pages of newspapers containing the word [3]. First, we collected the entire pages of the newspapers containing the word "coolie". Then, we extracted the text from the pages and searched the exact match for the word "coolie". This additional step ensures excluding false positive cases due to mis-recognized words. For instance, the search included the result of "cooli" even though it was not identical keyword that we wanted to query. In *New York Daily Tribune* published on August 5th, 1862 contained the word "cooli" but it accompanied many OCR errors making it doubtful whether the word "cooli" was actually from the word "coolie." [4] In order to reduce this kind of false positive cases, we double-processed the data by finding the exact match for the word "coolie" in the extracted text. Instead of finding a sentence that contains the word "coolie", we extracted upto ten tokens before the word "coolie" appeared and upto ten tokens after the word "coolie" appeared to create a pseudo-sentence. As some digitization of the newspapers were not perfect and thus missed punctuation marks, sentence tokenizer could not identify the sentence boundary correctly. This additional step of creating pseudo-sentence resulted in 125,253 text data (pseudo-sentence) containing the word "coolie" for the analysis.

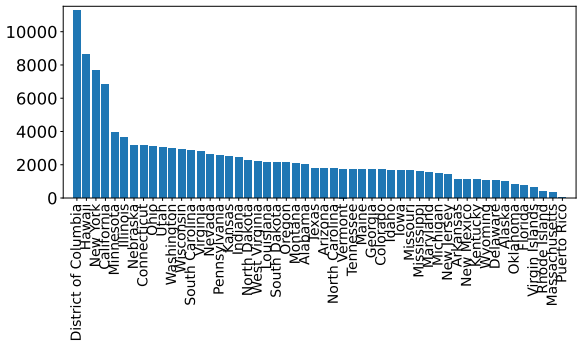

Figure 1: The count of text data containing the word "coolie" by State

In figure 1, we present the count of text data containing the word "coolie" by State. The count of text data is not evenly distributed across the States due to different digitization process of the newspapers. The most text data was from District of Columbia (*n=11,302*) followed by Hawaii (*n=8,613*) and New York (*n=7,671*). Puerto Rico (*n=15*), Massachusetts (*n=305*), Rhod Island (*n=418*), and Virgin Islands (*n=656*) had the least text data.

3.2 Data pre-processing

We pre-processed the data by removing the punctuation, non-alphabet tokens that might have been mis-recognized during the OCR process, and stop words. The list of stop words is from NLTK package in Python and we further converted the words into lemmas using Spacy [5]. Before we feed the data into the word embedding model, we ran the FastText model (Bojanowski et al., 2017) to identify possible OCR errors. FastText embedding takes character n-gram as input and outputs the embedding vector. This model was tested effective that it can generate possible OCR error candidates (Hajiali et al., 2022). By training entire sentence that contains the word "coolie", we identified 200 most similar words to the word "coolie". For instance, "coolieize" (0.8654), "oroolie" (0.8630), and "roolie" (0.8541) ranked high in the list of similar words to "coolie" according to the FastText embedding model. With 200 most similar words, we changed the top 200 words in the text data into

[2] https://chroniclingamerica.loc.gov/

[3] The data is collected on September 5th, 2023.

[4] https://chroniclingamerica.loc.gov/lccn/sn83030213/1862-08-05/ed-1/seq-4/ocr/ was searched as a page that contains "coolie" but it was due to the OCR error.

[5] https://spacy.io/

"coolie" and trained the Word2vec model in section 3.3.

3.3 RQ1. Word embedding

In order to answer RQ 1, we trained the Word2vec model (Mikolov et al., 2013) to use Continuous Bags of Words (CBOW) approach and the skip-gram approach. Both CBOW and skip-gram approach find the word embedding by predicting the target word from the context words. We trained the word2vec model with minimum word count of 5 and window size of 5 to generate the word embedding. We then took the average of the word embedding vector of the word "coolie" in each state and calculated the cosine similarity.

3.4 RQ2. Statistically over-represented words

In answering RQ 2, we grouped the newspapers into two groups: the newspapers from the then-Confederate States and the newspapers from the rest of the United States. For the then-Confederate States, we included the newspapers from Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, Virginia, and West Virginia. For the group for the rest of the United States, we included the newspapers from the rest of the states. We excluded sentences from the newspapers newspapers located in Virgin Islands (*n=656*) and Puerto Rico (*n=15*).

We calculated the log-odds ratio with informative Dirichlet prior (Monroe et al., 2008) by comparing the word frequency in the then-Confederacy newspapers and the newspapers published in the rest of the United States. The detailed metric is provided in equation 1.

$$\delta_w^{(i-j)} = \log \frac{y_w^i + a_w}{n^i + a_0 + y_w^i - a_w} - \log \frac{y_w^i + a_w}{n^i + a_0 - y_w^i - a_w} \quad (1)$$

The log-odds ratio with informative Dirichlet of each word w between two corpora i and j (in our study, newspapers from the then-Confederate States and the rest) given the prior frequencies are obtained from the entire corpus a. We selected 15,000 most frequent words from the entire corpus and the Z-score is calculated for each word. When n^i is the total number of words in corpus i, y_w^i is the number of times word w appears in corpus i, a_0 is the size of the corpus a, and a_w is the frequency of word w in corpus a (Kwak et al., 2020). With the log-odds ratio, we can identify the words that are over-represented in the corpora.

3.5 RQ 3. Text reprint detection

4 Results

4.1 RQ1. Comparing the meaning of "coolie"

In figure 2, we present the heatmap of cosine similarity comparison across the average embedding vector of the word "coolie" in each state. We are visually informed that the cosine similarity in most of the States is high except for a few States that created a semantically different meaning of the word "coolie". For instance, Massachusetts and Rhode Island showed average cosine similarity of 0.08 and 0.12, respectively when average cosine similarity across the entire States was 0.65. The most semantically dissimilar State to Massachusetts was when compared to Oklahoma (-0.10) and the most similar State to Massachusetts was North Dakota (0.23). In the meantime, Rhode Island showed the most dissimilar meaning of the word "coolie" when compared to Delaware (-0.03) and the most similar State to Rhode Island was Mississippi (0.23). Some then-Confederate States showed lower cosine similarity than the average cosine similarity across the entire States. For instance, Arkansas (0.43), Florida (0.48), and Tennessee (0.63) showed lower cosine similarity on average. The most dissimilar State to Arkansas was Massachusetts (-0.06) and the most similar State to Arkansas was Colorado (0.54). For Florida, the most dissimilar State was Rhode Island (0.06) and the most similar State to Florida was Nevada and Utah (0.61). For Tennessee, the most dissimilar State was Massachusetts (-0.03) and the most similar State to Tennessee was Utah (0.80).

To delve into the semantic difference of the word "coolie" in each State, we present the top 10 most similar words to the word "coolie" in the top 5 States that showed the most similar meaning of the word "coolie" and the top 5 States that showed the most dissimilar meaning of the word "coolie" in table 1 and table 2.

In table 1, we can observe that the word "coolie" was used in the context of labor, China, and wage. The word related to labor ("labor" and "work") appeared in the semantically close words in Illinois, Wisconsin, Virginia, and Nevada. "chinese" was the most similar word of identification of coolie's ethnicity (Illinois, Wisconsin, Virginia, and Nevada) while "japanese" appeared in Wisconsin

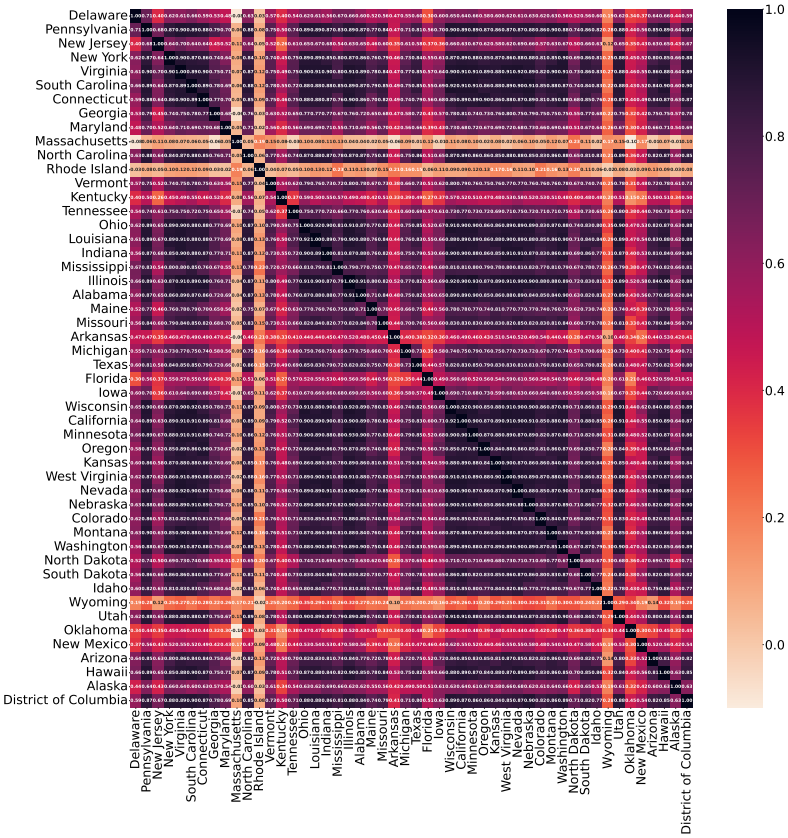

Figure 2: The heatmap of cosine similarity comparison across the average embedding vector of the word "coolie" in each state

The highest five States				
Illinois	California	Wisconsin	Virginia	Nevada
labor (0.9998)	country (0.9995)	chinese (0.9998)	chinese (0.9998)	chinese (0.9997)
chinese (0.9998)	bill (0.9995)	labor (0.9998)	man (0.9997)	club (0.9997)
wage (0.9997)	upon (0.9995)	two (0.9998)	trade (0.9997)	labor (0.9997)
two (0.9997)	well (0.9995)	one (0.9998)	work (0.9997)	make (0.9997)
one (0.9997)	go (0.9995)	time (0.9998)	one (0.9997)	use (0.9996)
day (0.9997)	stop (0.9995)	carry (0.9997)	three (0.9997)	say (0.9996)
china (0.9997)	many (0.9995)	take (0.9997)	number (0.9997)	importation (0.9996)
man (0.9997)	american (0.9995)	make (0.9997)	make (0.9997)	man (0.9996)
pay (0.9997)	con (0.9995)	japanese (0.9997)	importation (0.9997)	trade (0.9996)
say (0.9997)	would (0.9995)	would (0.9997)	two (0.9997)	day (0.9996)

Table 1: Top 10 most similar words to the word "coolie" in the top 5 States that showed the most similar meaning of the word "coolie"

as well. The word related to wage ("wage", "pay") appeared in the high closest word to "coolie."

Among the highest five States, there are many common words across the States that might have created an embedding for the word "coolie" not so much different from other States. Words like "make", "say", "man", and/or numbers like "one" and "two" are common words across the highest five States.

However, the most dissimilar States, Massachusetts and Rhode Island, showed the different context of the word "coolie" as presented in table 2. Unlike the highest five States, where the word "coolie" was used in the context of labor, China, and wage, Massachusetts and Rhode Island created a unique discourse of the word "coolie." For instance, the word "labor" and "work" does not appear in the top 10 most similar words to the word "coolie" in Massachusetts and Rhode Island. Similarly, we cannot find "chinese" or "japanese" that refer to the ethnicity of coolie workers in the top 10 most similar words as well as "wage" or "pay." Instead, Massachusetts has "among", "call", "know", "prohibit", "report", "get", "arrive", and "come" as semantically close words to "coolie." However, these words are not found in top 10 closest words in the highest five States in table 1. A unique embedding

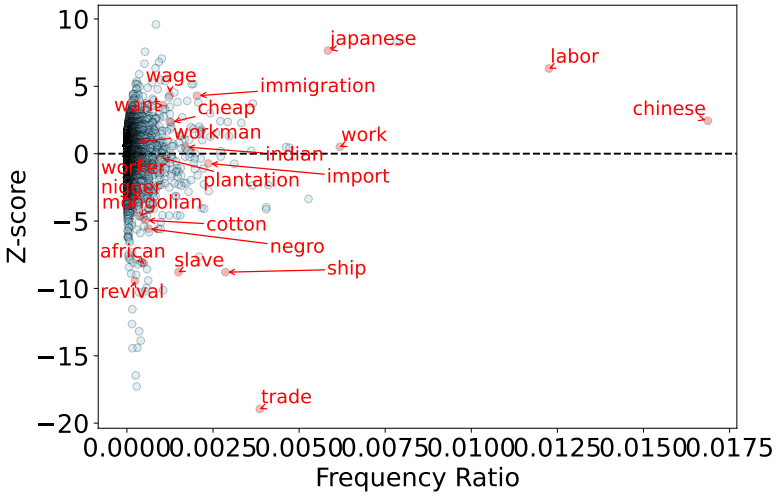

Figure 3: The Z-score of words in the newspapers

The lowest five States				
Massachusetts	Rhode Island	Wyoming	Oklahoma	Arkansas
among (0.3579)	order (0.4116)	chinese (0.9969)	chinese (0.9821)	labor (0.9985)
call (0.3201)	india (0.3972)	labor (0.9969)	shoulder (0.9815)	chinese (0.9984)
know (0.2832)	woman (0.3922)	would (0.9955)	japanese (0.9776)	mongolian (0.9978)
prohibit (0.2295)	great (0.3890)	japanese (0.9952)	pay (0.9748)	one (0.9978)
time (0.2232)	take (0.3761)	six (0.9950)	also (0.9725)	thousand (0.9977)
report (0.2221)	ship (0.3745)	one (0.9949)	labor (0.9706)	japanese (0.9975)
get (0.2209)	law (0.3432)	bring (0.9948)	carry (0.9632)	tolerate (0.9975)
arrive (0.2176)	united (0.3260)	say (0.9946)	home (0.9632)	revivial (0.9974)
come (0.2131)	carry (0.3234)	work (0.9944)	get (0.9630)	may (0.9974)
two (0.2104)	many (0.3122)	two (0.9941)	work (0.9612)	carry (0.9974)

Table 2: Top 10 most similar words to the word "coolie" in the top 5 States that showed the most dissimilar meaning of the word "coolie"

for the word "coolie" in Rhode Island could also be attributed to the unique discourse created by uncommon words such as "order", "india", "woman", "great", "ship", "law", "united", and "many." However, Wyoming, Oklahoma, and Arkansas share common words with the States in table 1. The words like "chinese", "labor", "would", "japanese", "one", "say", "work" in Wyoming are among top 10 similar words in the highest five States. Despite existence of common words, "shoulder", and "home" in Oklahoma and "mongolian", "tolerate", and "revival" in Arkansas are unique words that are not found in the top 10 similar words in the highest five States.

4.2 RQ 2. Statistically over-represented words in the South and the rest of the United States

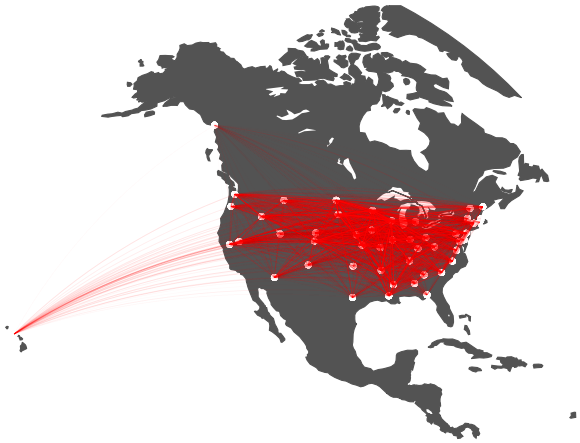

Figure 4: The Z-score of words in the newspapers

4.3 Discussion

5 Limitations and Future Work

References

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A Example Appendix